

Explorative Datenanalyse

zur Vorbereitung einer Klassifikation
(Machine Learning)

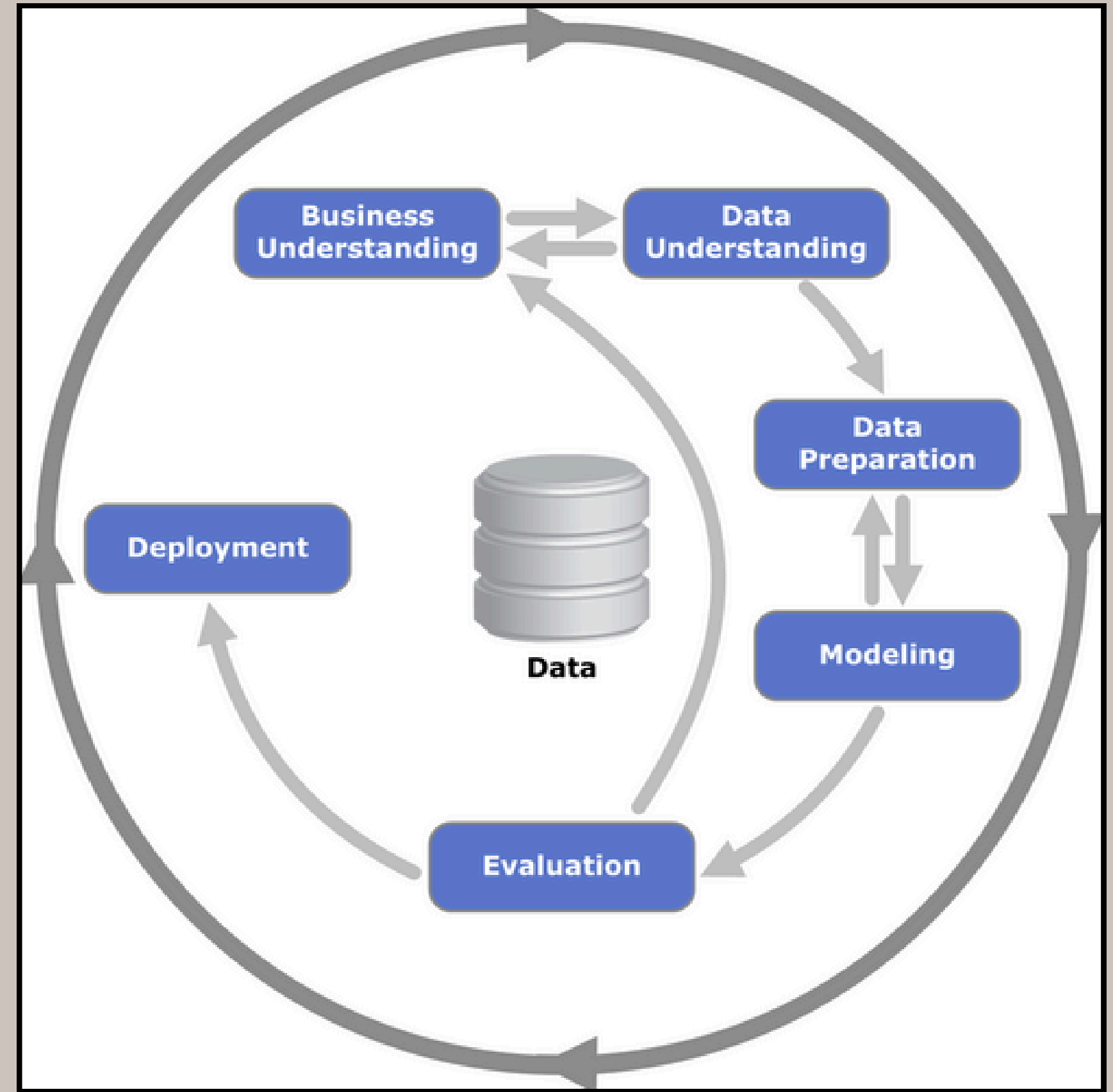
mit Nutzerdaten eines
Sportuhrenherstellers

Use Case

Gezielte Kundenansprache Predictive Analytics

Was bewegt Kunden dazu, das neueste
Modell zu kaufen?

Welche potenziellen Kundengruppen
können identifiziert werden?



Quelle: https://en.wikipedia.org/wiki/Cross-industry_standard_process_for_data_mining#/media/File:CRISP-DM_Process_Diagram.png, 8.3.2026

Jede Zeile eine Person - Wie könnten die aussehen?



Data Exploration - Features

Quantitative Performance

Swimming hours per week
Biking hours per week
Running hours per week
Total training hours

Target

User of the latest model

Personal information

Age
sex
country
town
Goal of training
Preferred Training Daytime
Color of watch

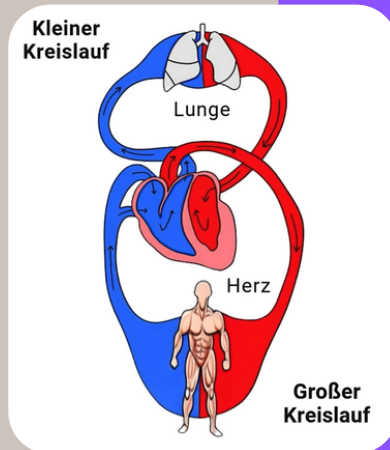
Qualitative Performance

VO2 MAX
10k Running Prediction
Calories burned per week

ID

Technical information

Support cases
Customer years
Most current software update
Subscription type
Synchronisation



Aufgabe und Schnittstellen

1. Data Exploration

Datenqualität & logistische Regression

2. Data Preparation and Model Baseline

Decision Trees, ANN, SVM, ...

3. Modeling Optimization

Klassifikation - Modelle und Bedingungen

Models and Conditions

Logistic Regression

explainable
NaN handling required
scaling recommended
outlier treatment important
multicollinearat

Decision Trees, XG Boost, Random Forest

explainable
NaN handling required
no scaling
no outlier treatment

Naive Bayes, Neural Networks

Blackbox
probabilities
'naive' = independent features
amount of data

Weiteres Vorgehen im
Projekt: Baseline Logistische
Regression

Distance-based models

k-means
Clustering

SVM

PCA

k-NN

Conditions

Scaling
categorical variables - OHE
Outlier treatment

Curse of dimensionality

Feature selection

Gesundheit

Lifestyle

Community

Motivation

Wettkampf

Kontrolle



Bildquelle: pixabay, cc



Theorien

Beispiele zufällig

Non-User

=====

STATUS: LEGACY MODEL (NON-ADOPTER)

=====

ID: C0633

Gender: Other

Age: 46 years

Location: Melbourne, Australia

Training Goal:Competition

Weekly Hours: 18.9 hrs

Fitness Level: VO2 Max 63.0

Subscription: Free

Watch Style: Black Case



User



=====

PERSONA B: THE LATEST ADOPTER

=====

ID: C0503

Gender: Other

Age: 21 years

Location: New York, USA

Training Goal:Competition

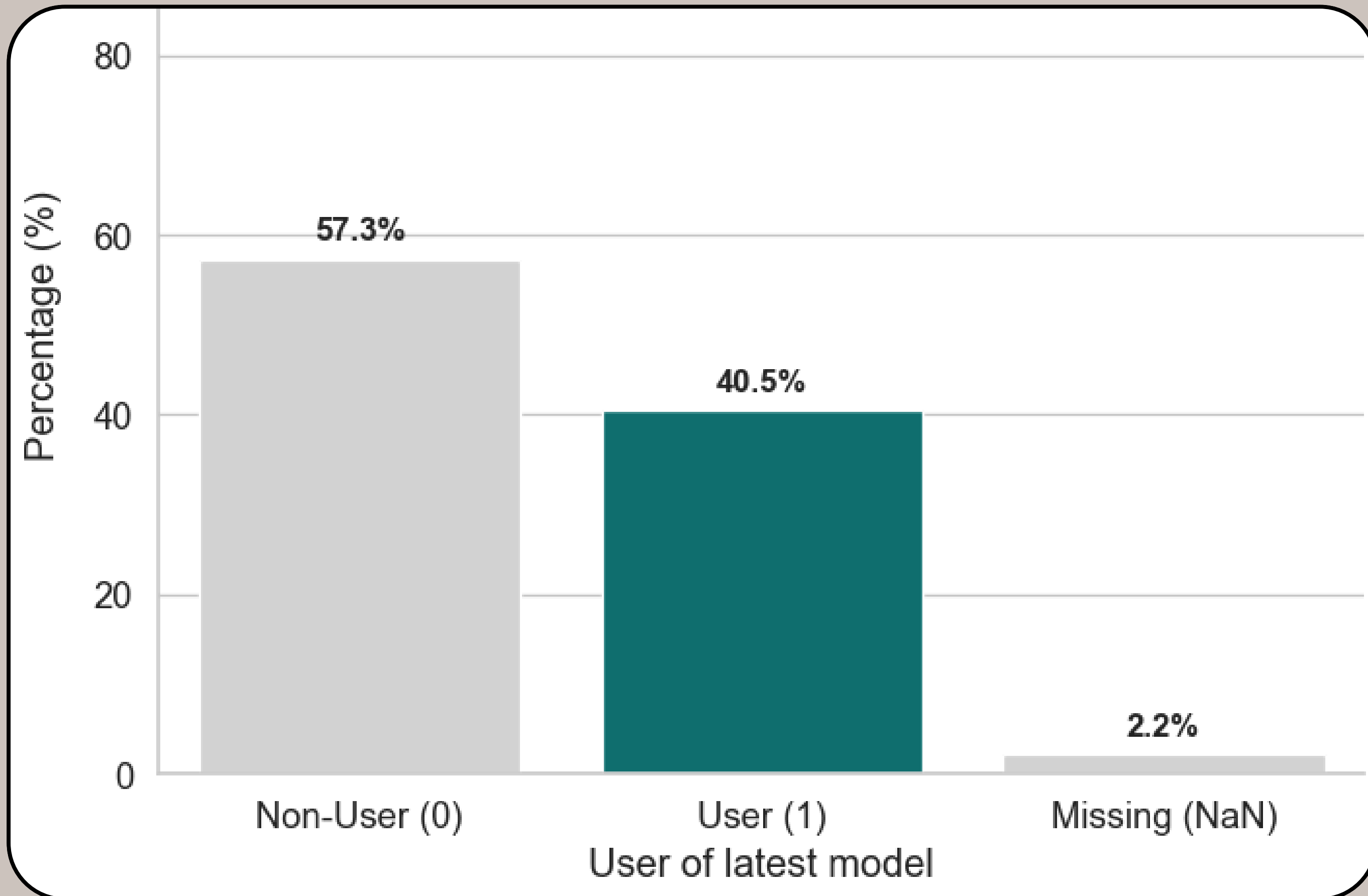
Weekly Hours: 12.1 hrs

Fitness Level: VO2 Max 56.2

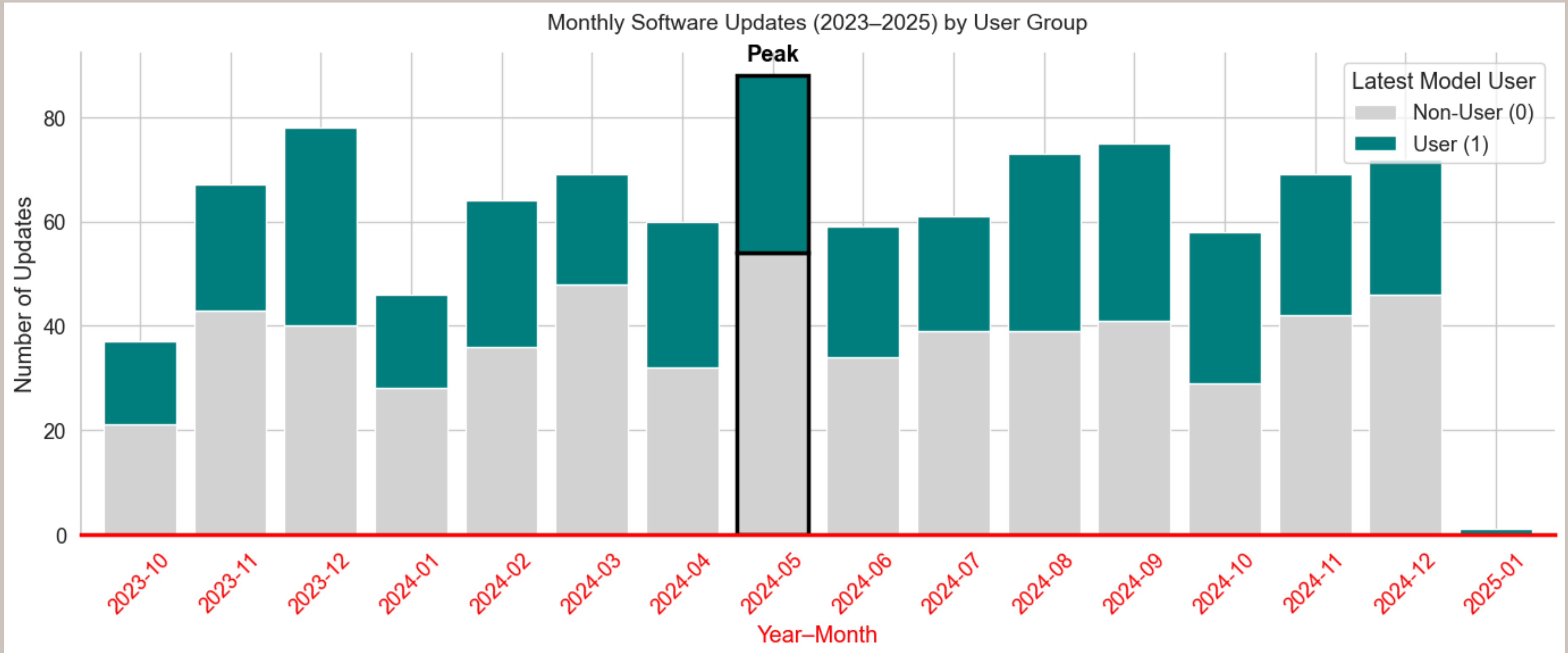
Subscription: Free

Style: Black Watch

Zielvariable und Baseline Model



Zeitraahmen und Barplots

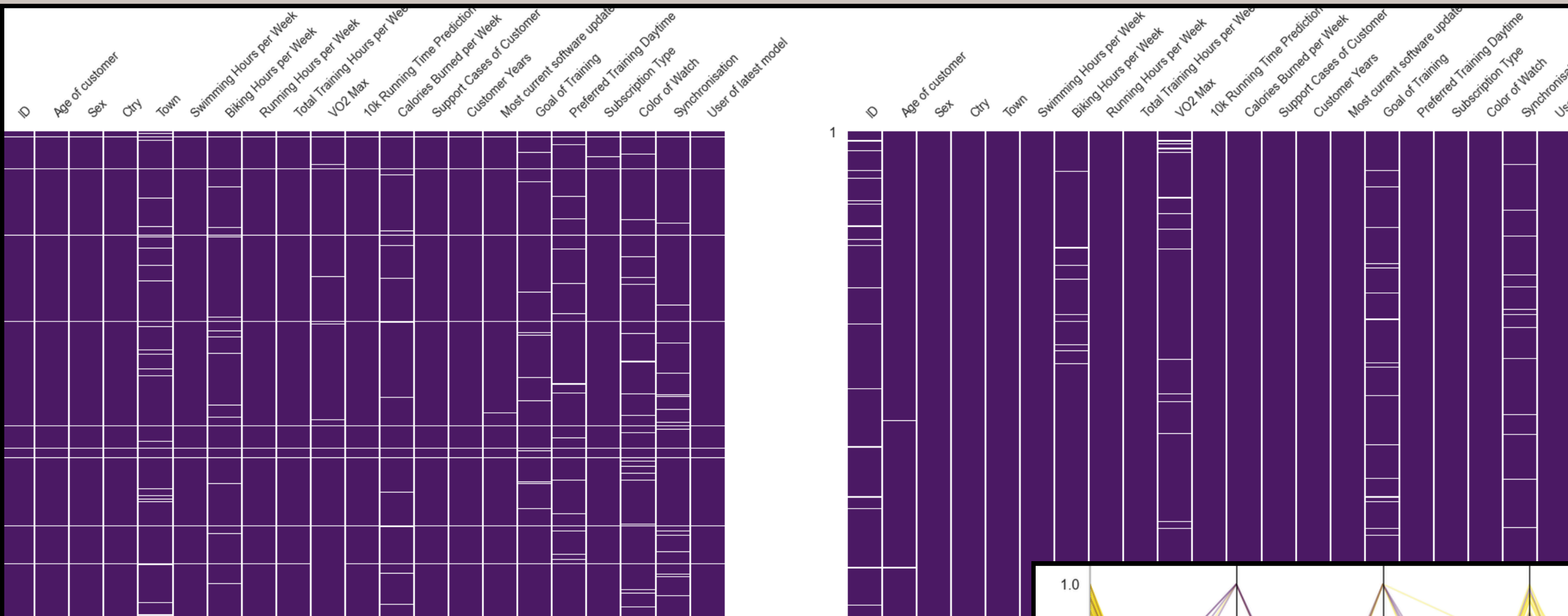


Feature Engineering: Date to days

Zeitspanne

Datenqualität

Fehlende Werte



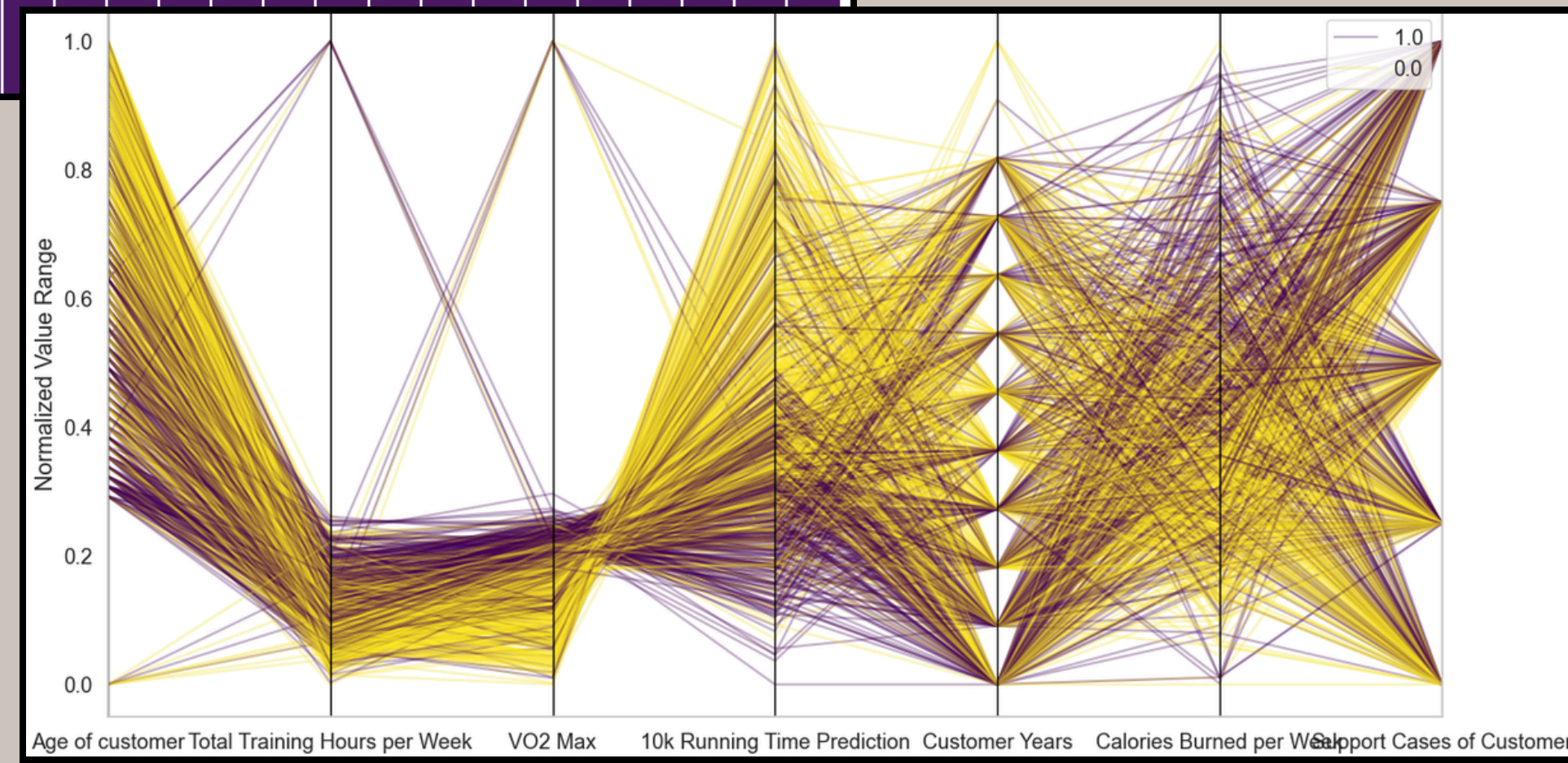
Duplikate
via ID

Ausreißer

IQB-Methode

StandardScaler

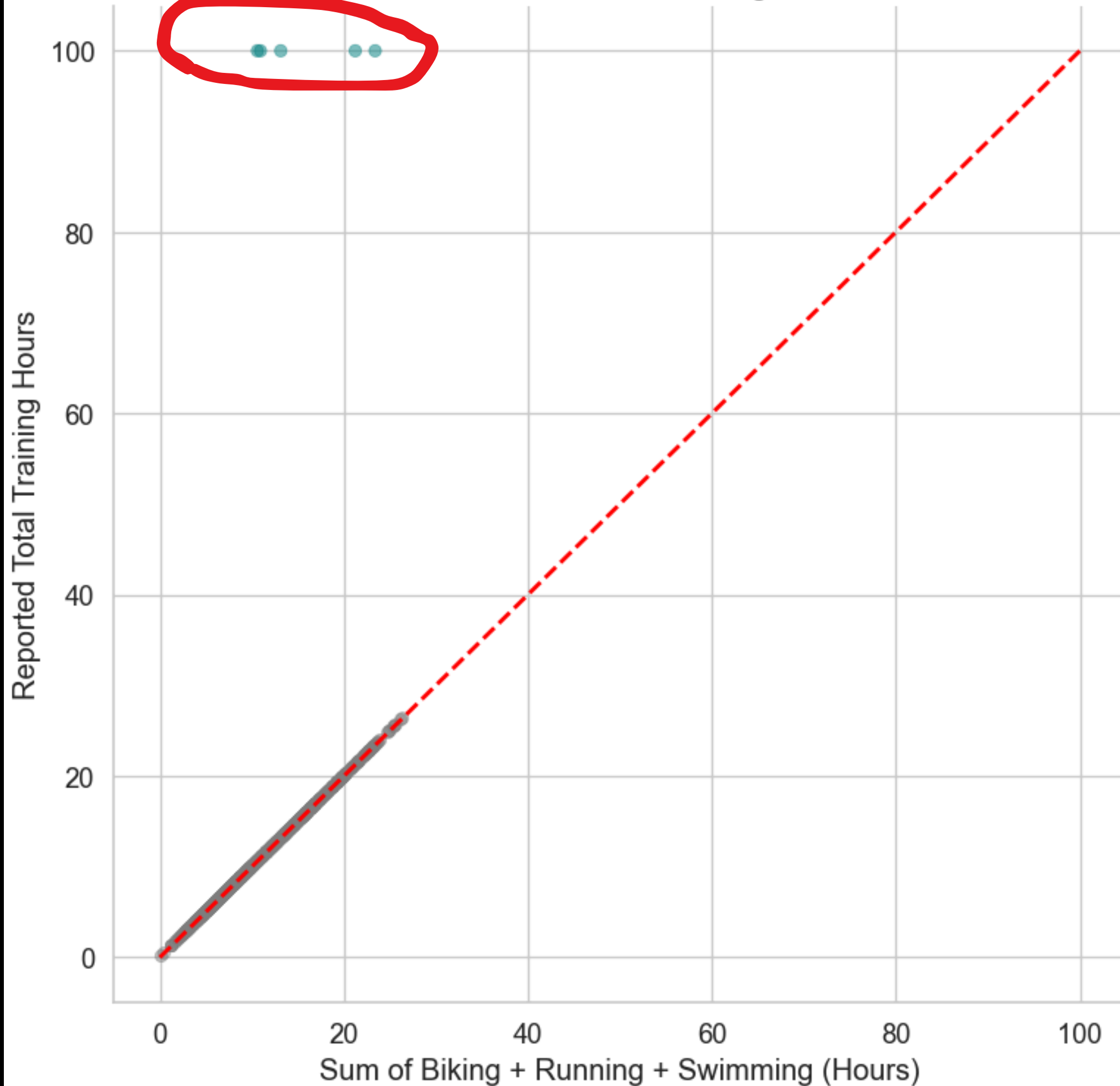
Winzorization



EDA: Deskriptive Statistik

	count	mean	std	min	25%	50%	75%	max
Age of customer	1476.0	40.36	14.03	-1.00	29.00	41.00	52.00	64.00
Swimming Hours per Week	1478.0	2.21	1.59	0.00	0.99	1.98	3.20	9.97
Biking Hours per Week	1444.0	5.51	3.46	0.00	2.70	5.18	7.70	16.11
Running Hours per Week	1478.0	3.28	2.23	0.00	1.64	2.92	4.62	12.28
Total Training Hours per Week	1478.0	11.42	7.84	0.15	7.38	10.32	14.12	100.00
VO2 Max	1460.0	52.76	22.14	10.22	42.99	54.96	61.73	247.26
10k Running Time Prediction	1478.0	45.55	7.50	26.90	40.22	43.84	50.13	69.02
Calories Burned per Week	1459.0	5482.20	2494.14	-47.26	3670.61	5184.27	6994.62	13772.98
Support Cases of Customer	1478.0	1.98	1.41	0.00	1.00	2.00	3.00	4.00
Customer Years	1478.0	3.85	2.90	0.00	1.00	4.00	6.00	11.00
User of latest model	1478.0	0.41	0.49	0.00	0.00	0.00	1.00	1.00

Validation: Outliers in Training Hours

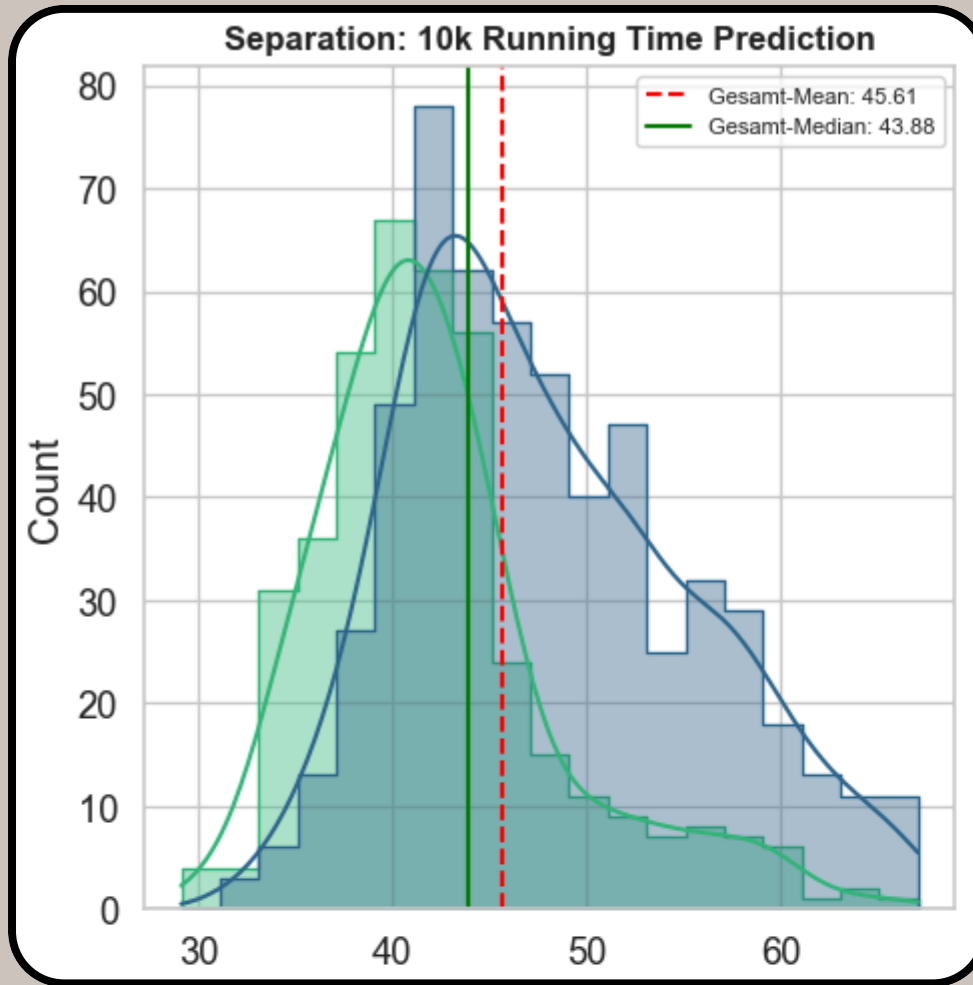


**Total sum of
training hours**

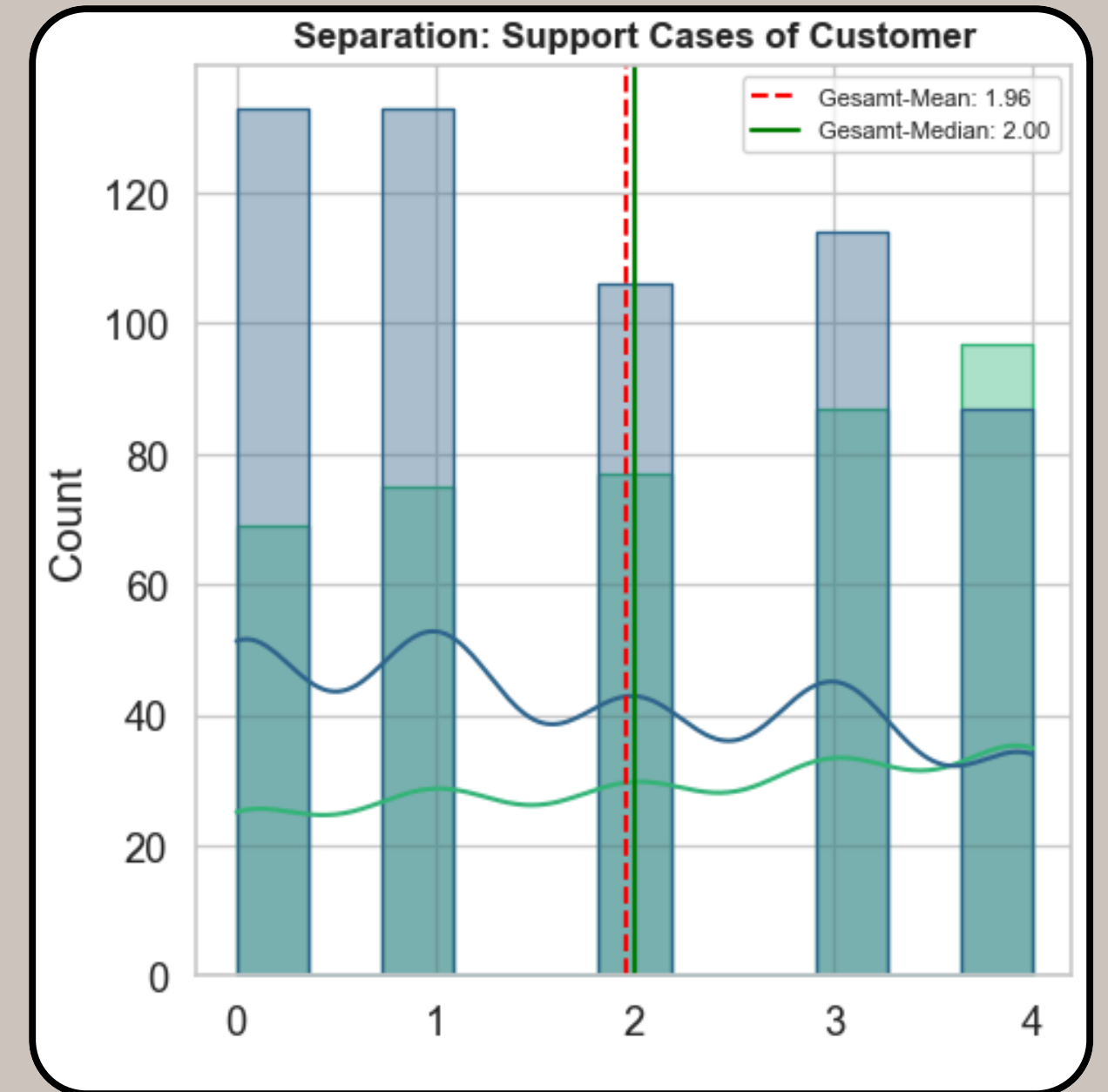
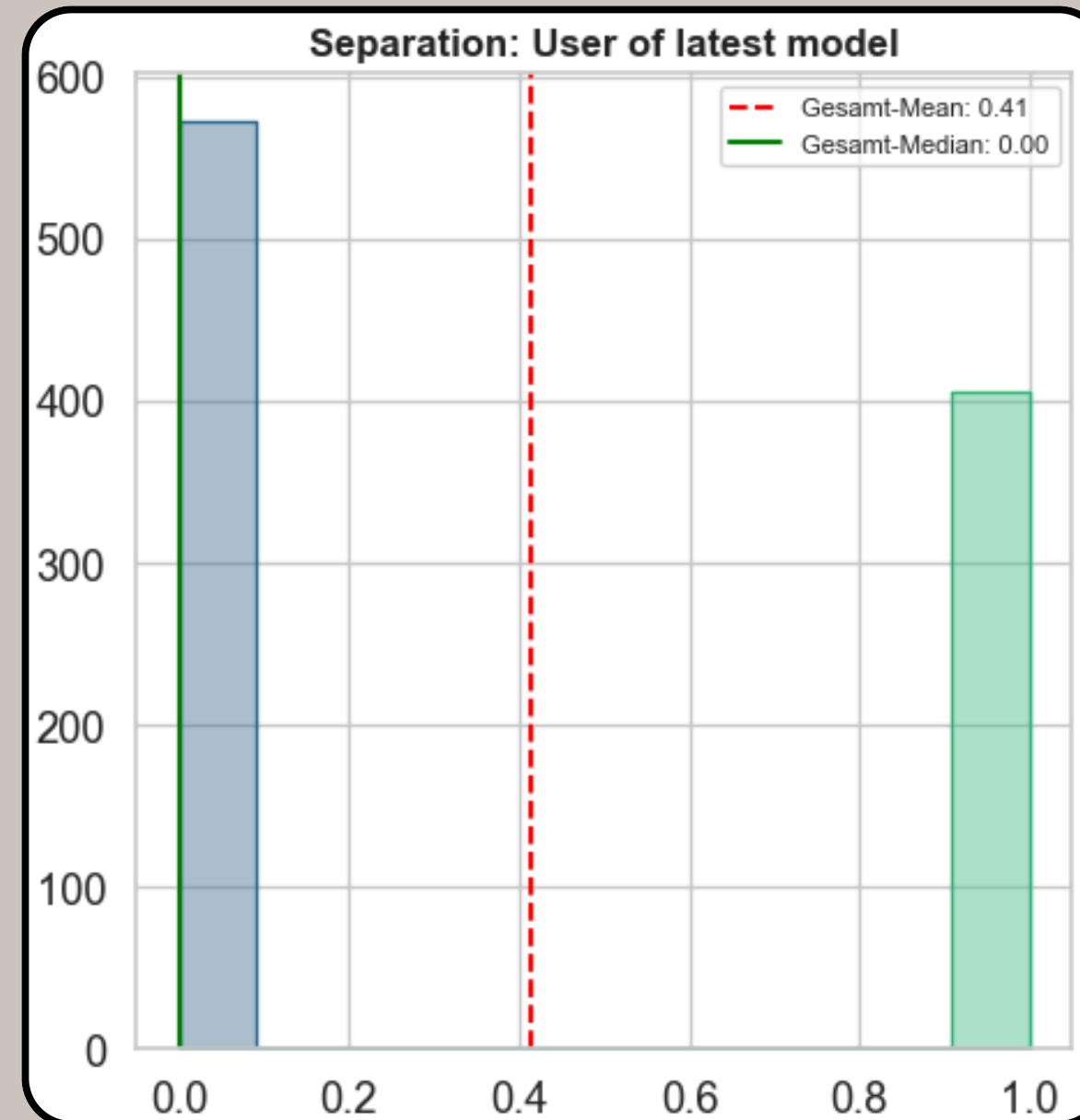
Multikollinearität

Summe teilweise
inkorrekt

Histogramme

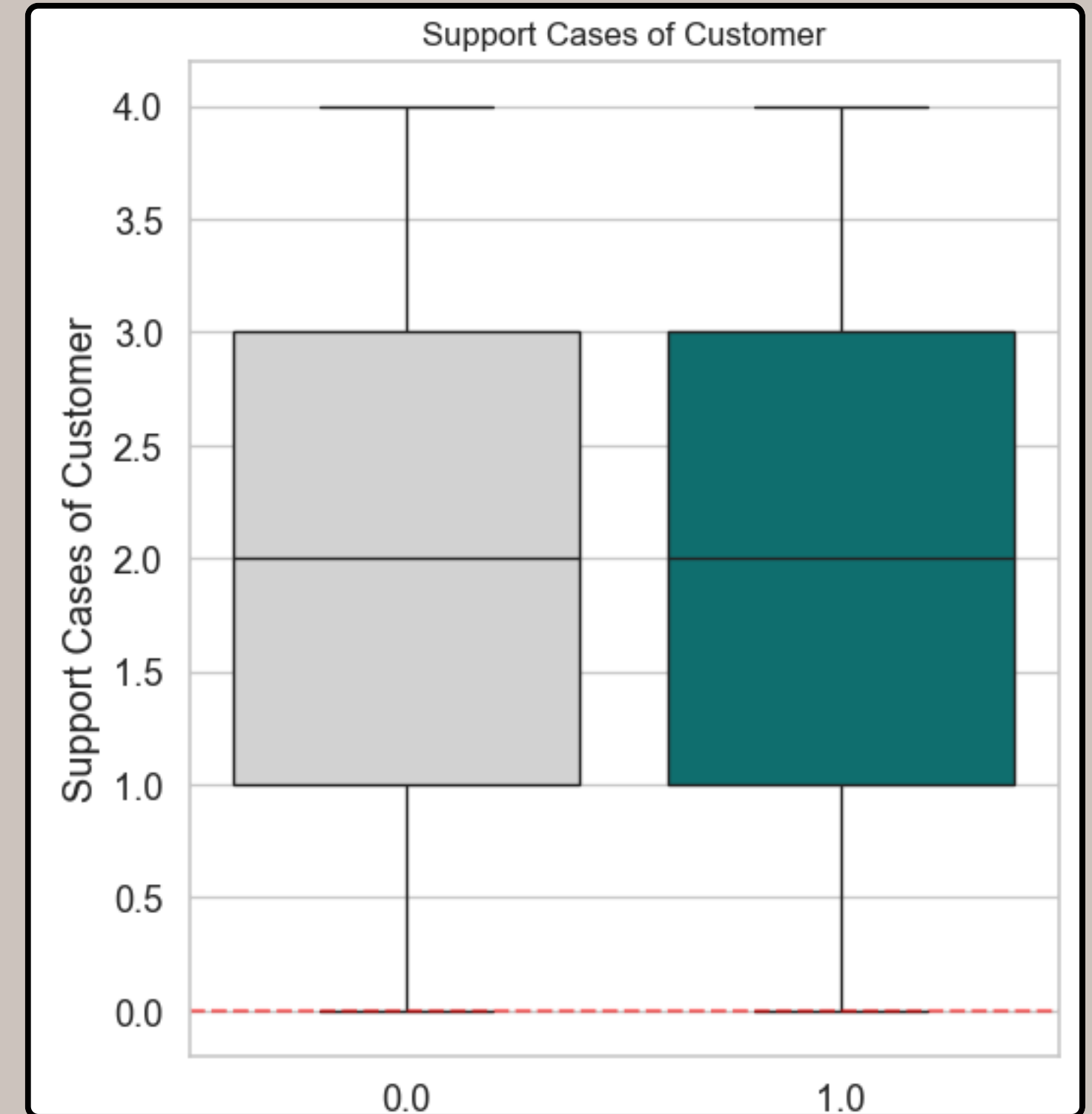
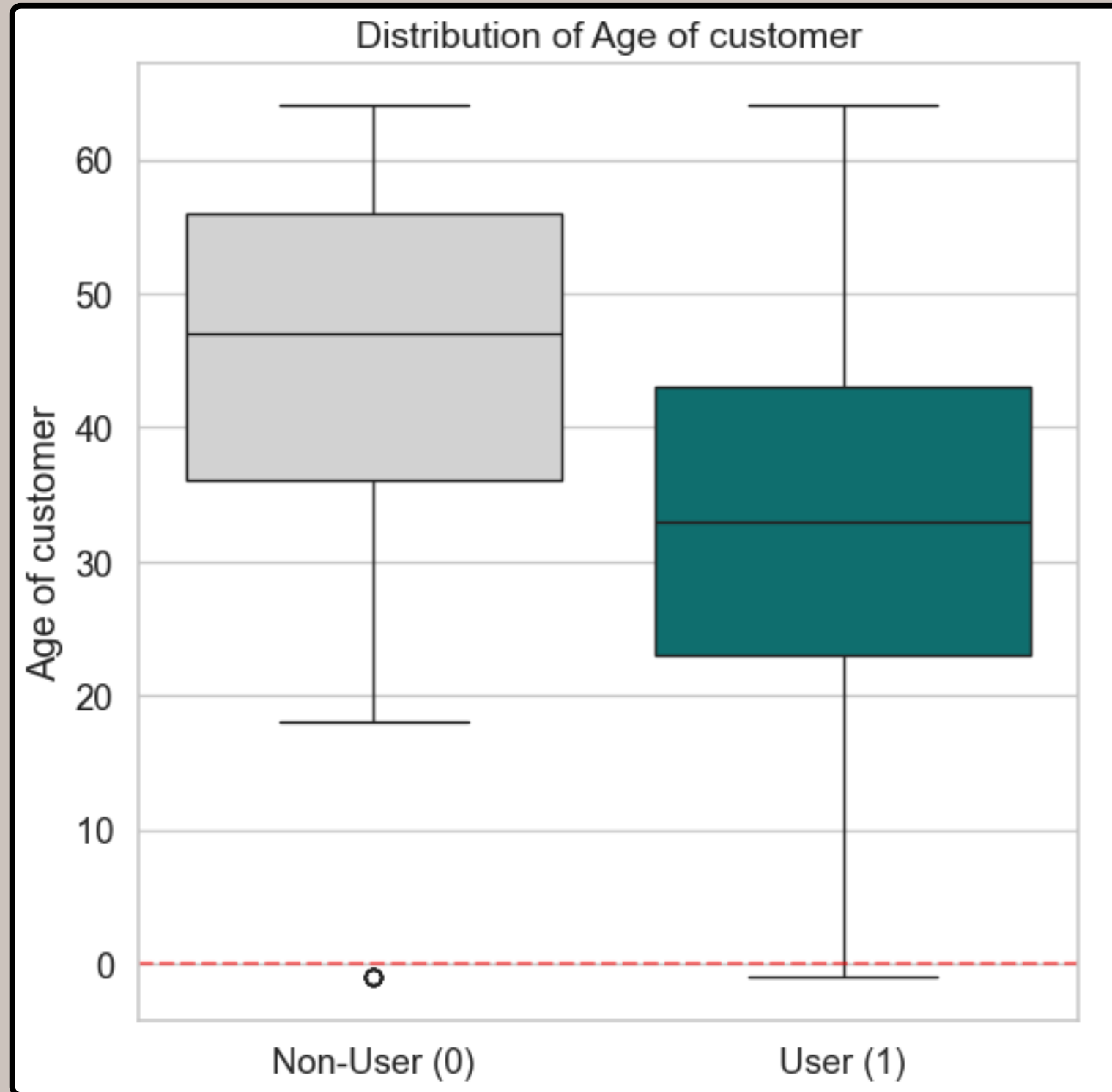


Trennung
& Schiefe

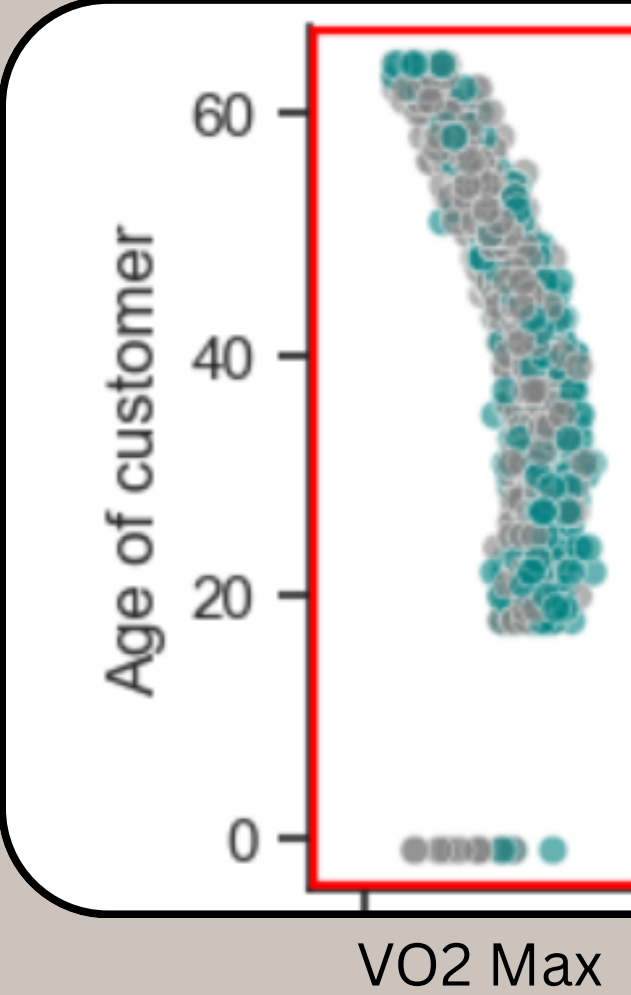
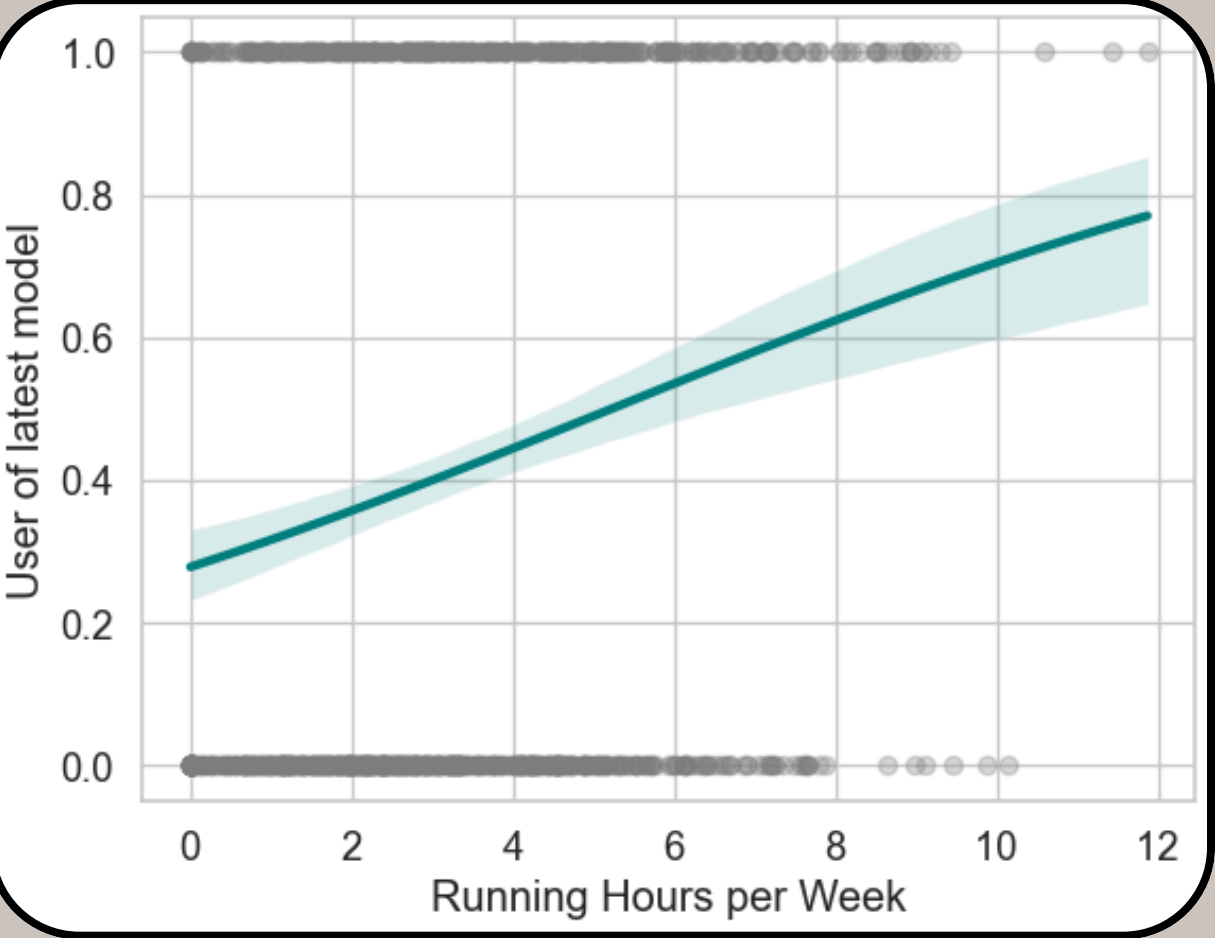
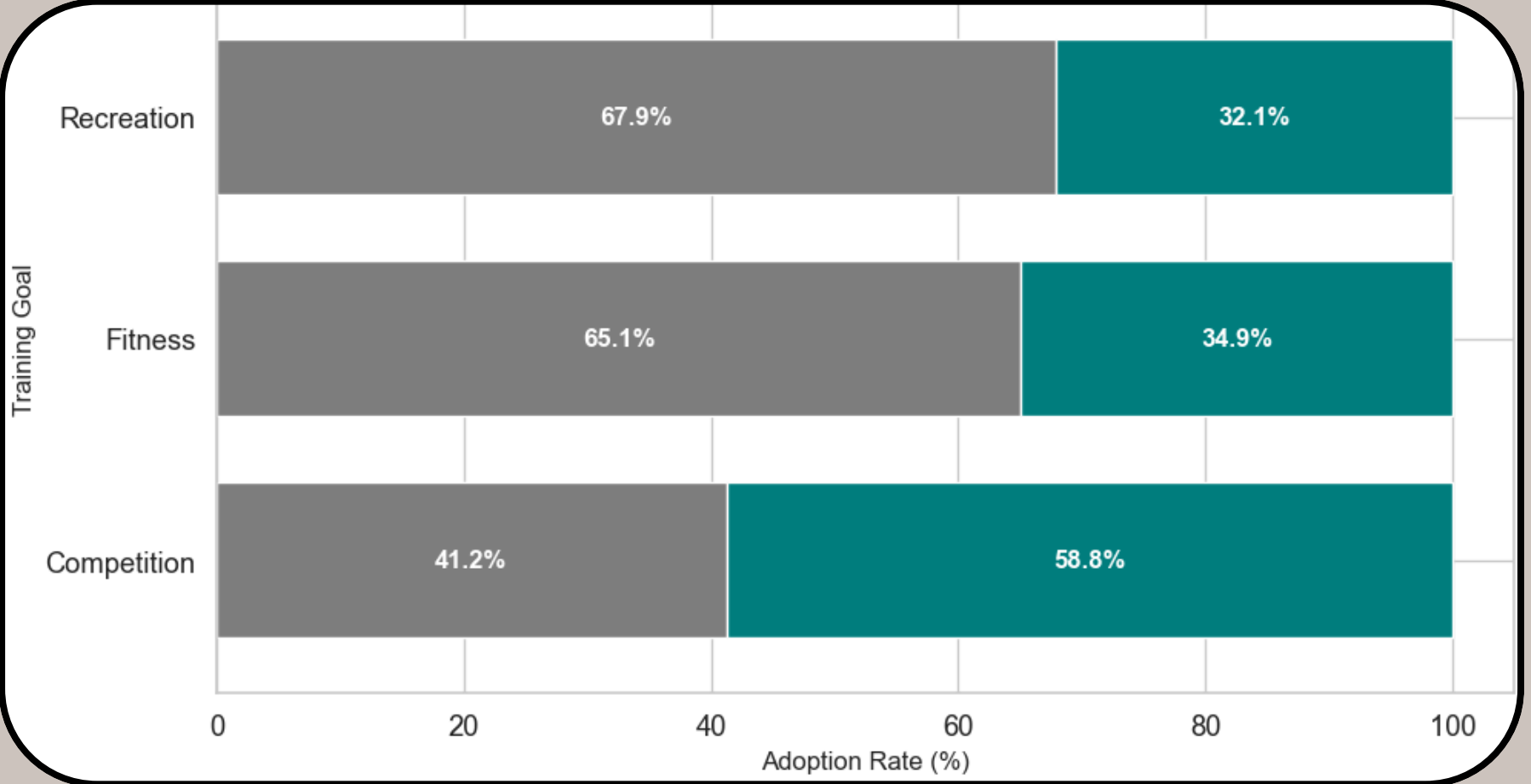
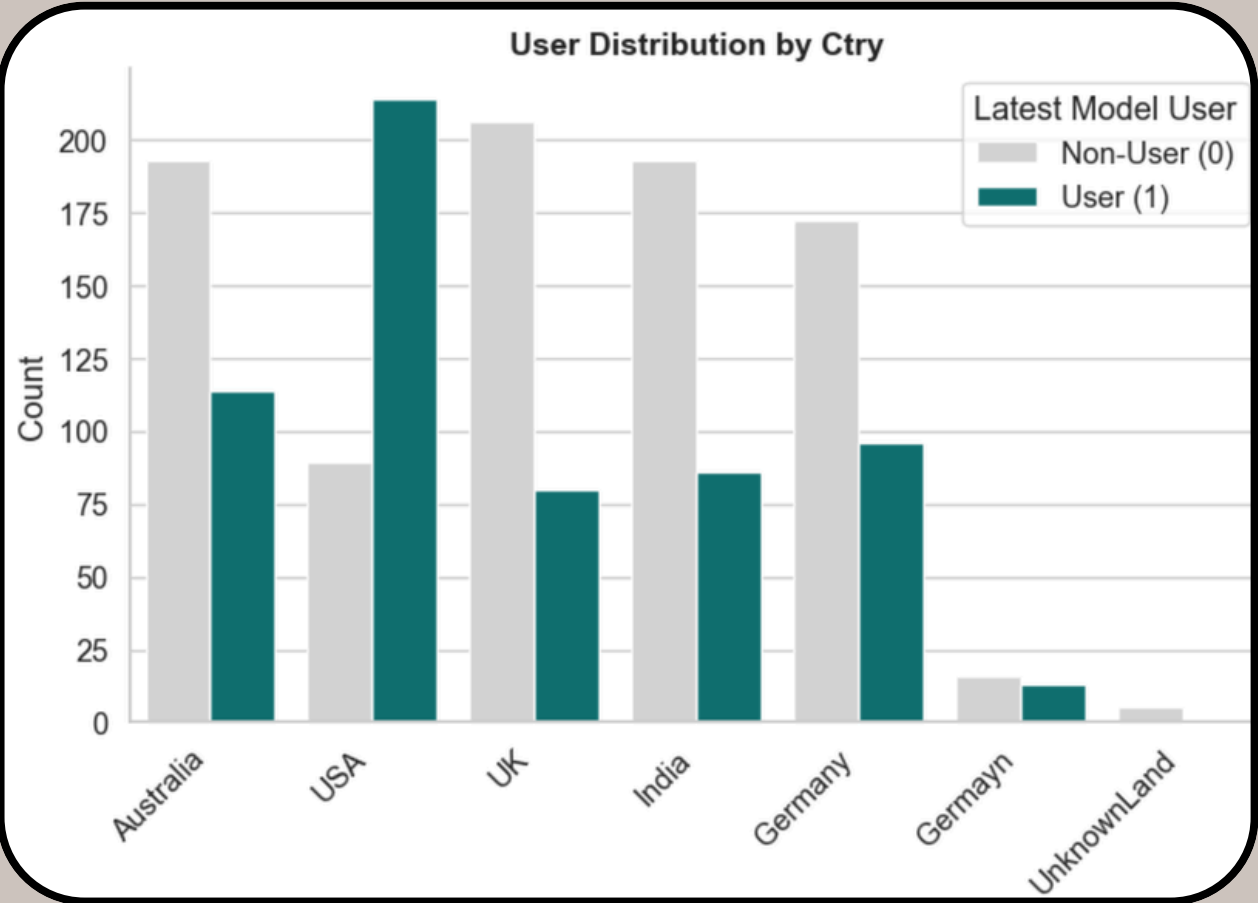
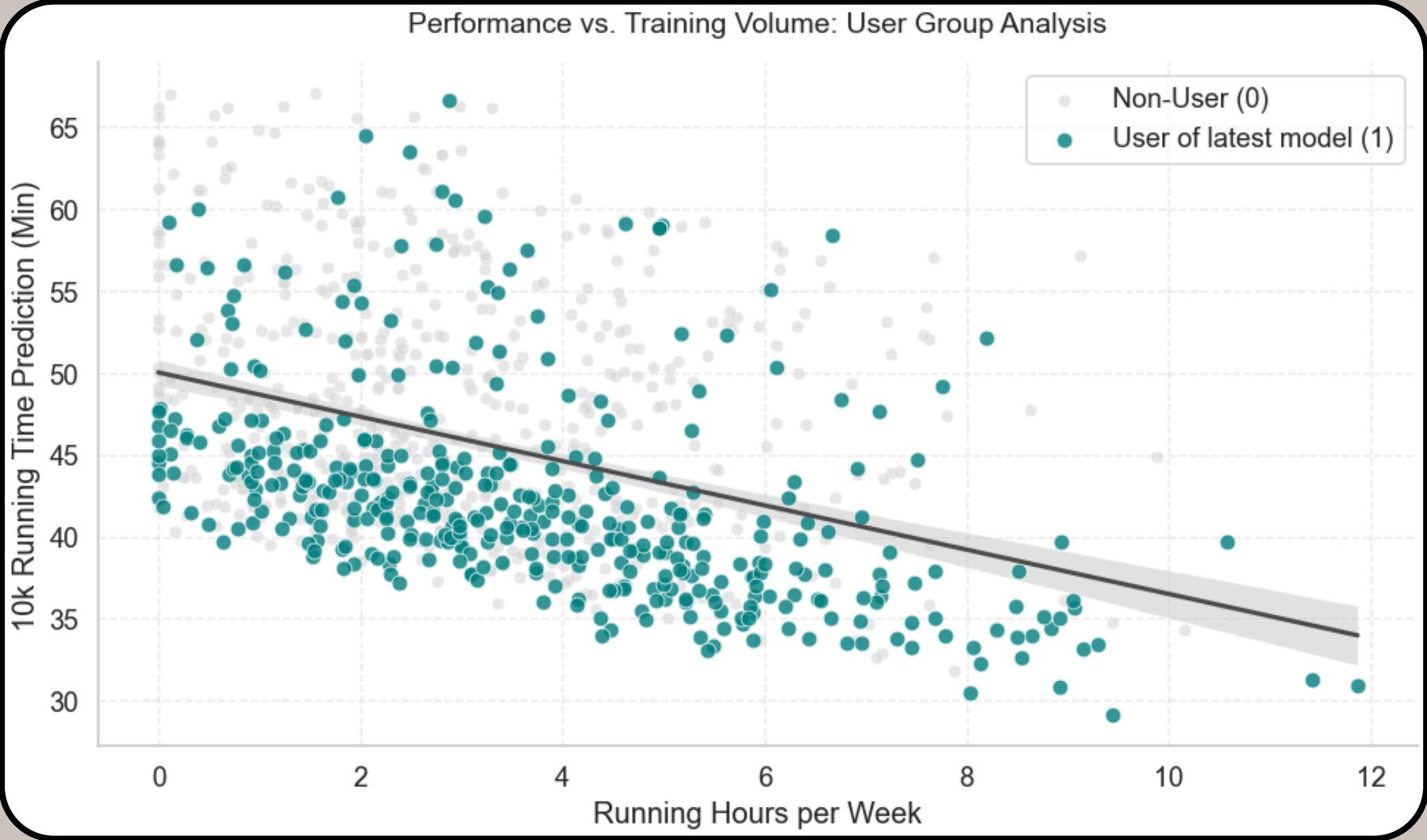


numerische Features

Kategoriale Variablen: Boxplots



Erkenntnisse ...



Univariate Analysis

Feature selection

VIF

Chi-Quadrat-Test

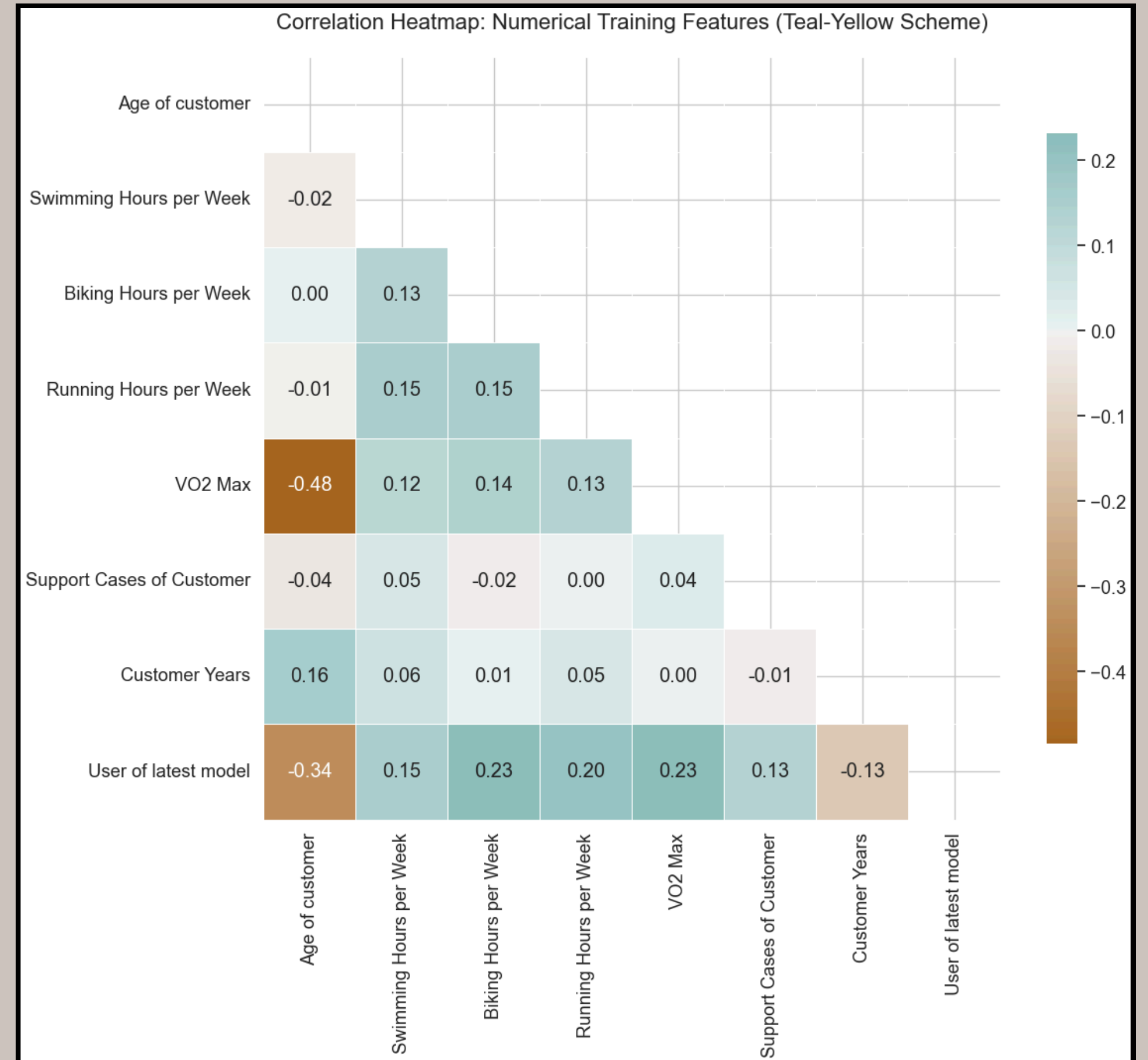
ANOVA

VIF

Feature	VIF
Calories Burned per Week	429.218770
Biking Hours per Week	180.257265
Running Hours per Week	64.966675
Swimming Hours per Week	33.658406
10k Running Time Prediction	31.492134
Total Training Hours per Week	24.461000
Age of customer	22.971881
VO2 Max	7.101454
Support Cases of Customer	2.916240
Customer Years	2.738743

Feature	VIF
VO2 Max	5.311504
Age of customer	5.086852
Biking Hours per Week	3.721766
Running Hours per Week	3.388353
Swimming Hours per Week	3.020206
Support Cases of Customer	2.762214
Customer Years	2.738662

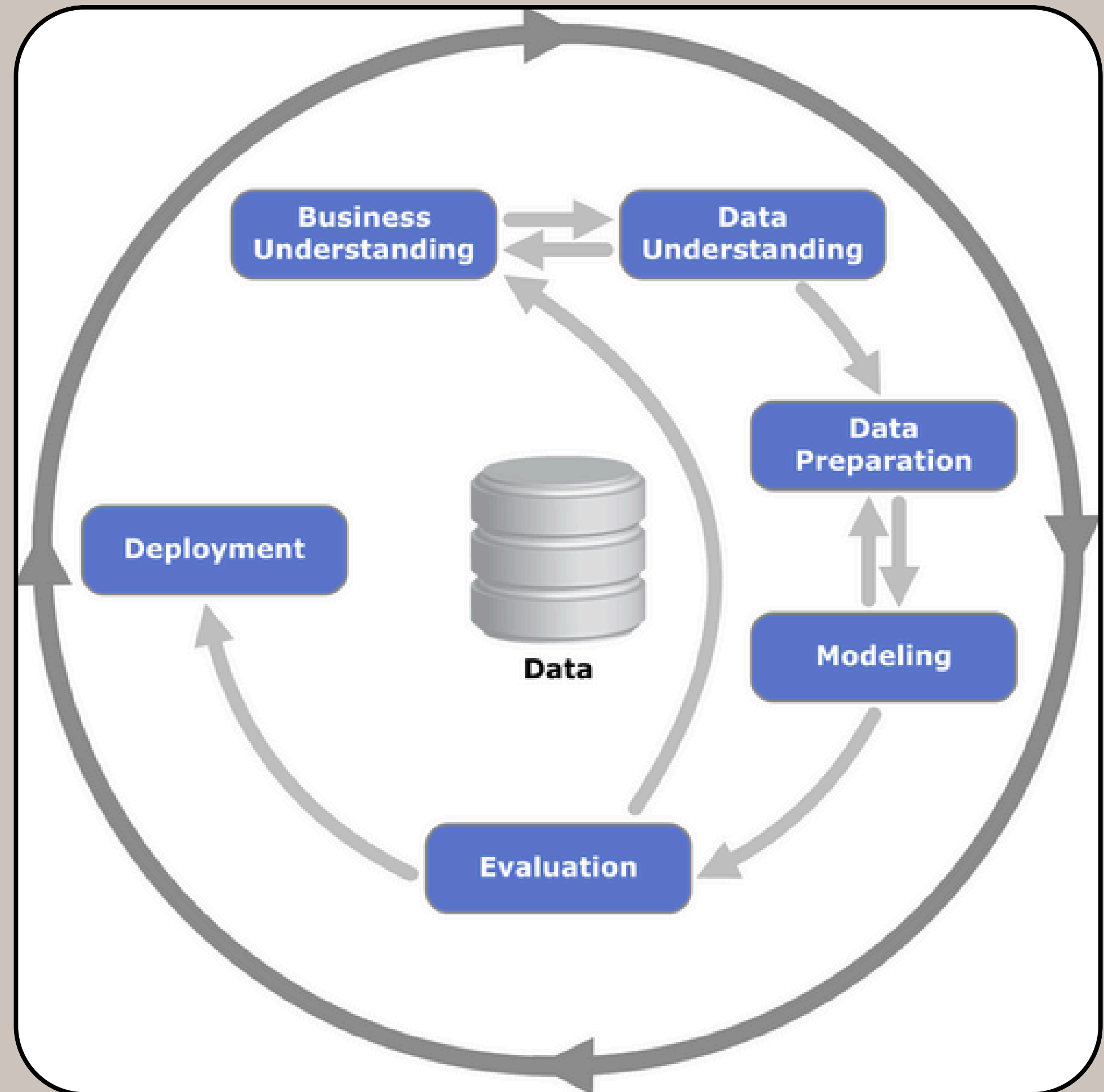
Linearität, Kausalität...



Übergabe

- Proposal for preprocessing
- Missing values
- Outliers
- Features to be transformed
- Feature selection
- Duplicates

Preprocessing und ML-Algorithmen



Quelle: https://en.wikipedia.org/wiki/Cross-industry_standard_process_for_data_mining#/media/File:CRISP-DM_Process_Diagram.png, 8.3.2026

Quellen

Literatur

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Weitere Quellen

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<https://www.youtube.com/watch?v=gP8SX9XnbhE>, 15.2.2026

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